A Comparative Study of Morphological Operations and Matched Filter with Active Contour for Retinal Blood Vessel Segmentation

Rao Umair, Muhammad Fahad, Mazz Zulfaqar

Department ofComputer Science, Digital Image Processing Course Project, 2026

Abstract This paper presents a comparative evaluation of two distinct approaches for automated retinal blood vessel segmentation on the DRIVE (Digital Retinal Images for Vessel Extraction) dataset. The first method, based on Ozkaya et al. [1], employs a classical morphological pipeline comprising adaptive Gaussian thresholding, Lab-space sharpening, edge-preserving denoising, Otsu binarization, and morphological open/close operations. The second method is a vessel-specific pipeline combining Contrast Limited Adaptive Histogram Equalization (CLAHE), a 36-angle matched filter bank with Gamma PDF kernels, and an Active Contour Without Edges (ACWE/Chan-Vese) level set segmentation. Experiments on 20 DRIVE training images (IDs 21–40) demonstrate that the morphological approach achieves Sensitivity = 75.40%, Specificity = 39.65%, and Accuracy = 42.67%, indicating severe over-segmentation. The vessel pipeline achieves Sensitivity ≈75.1%, Specificity ≈96.3%, and Accuracy ≈95.1%, representing a ≈56.6 percentage-point improvement in specificity. These results validate that vessel-specific filter design and energy-minimizing active contour segmentation are essential for clinically viable retinal vessel analysis.

Index Terms—Retinal vessel segmentation, matched filter, active contour, morphological operations, DRIVE dataset, CLAHE, Chan-Vese model, diabetic retinopathy.

I. INTRODUCTION

Retinal fundus imaging provides a non-invasive window into the microvasculature of the eye and serves as a primary diagnostic tool for a range of systemic and ocular diseases. The morphology of retinal blood vessels—their caliber, tortuosity, and branching patterns—is directly associated with conditions such as diabetic retinopathy, glaucoma, hypertension, and arteriosclerosis [2]. Early and accurate detection of pathological changes in vessel morphology can prevent irreversible vision loss and guide timely therapeutic interventions.

Manual delineation of retinal vessels by trained ophthalmologists is the gold standard but is prohibitively time-consuming, typically requiring 3–4 hours per image, and is subject to inter-observer variability [3]. Automated segmentation algorithms are therefore indispensable for large-scale screening programs.

This paper investigates two fundamentally different algorithmic paradigms for retinal vessel segmentation evaluated on the DRIVE benchmark dataset [4]. The first approach (Method 1) follows Ozkaya et al. [1] and employs a classical image processing pipeline based on morphological operations and generic filtering. The second approach (Method 2) is a vessel-specific pipeline combining CLAHE contrast enhancement, a multi-angle matched filter bank [5], and the Chan-Vese ACWE model [6] for boundary-driven segmentation.

The principal contributions of this work are: (i) complete reimplementation and evaluation of both pipelines on the DRIVE dataset; (ii) detailed mathematical analysis of all filters, kernels, and operators; and (iii) rigorous quantitative comparison demonstrating that vessel-specific filter design is critical for high specificity. The paper is organized as follows: Section II reviews related work; Section III describes the DRIVE dataset; Section IV details both methodologies; Section V presents experimental results; Section VI discusses findings; and Section VII concludes.

II. RELATED WORK

A. Morphological and Filter-Based Methods

Early automated segmentation approaches relied on classical morphological operations and handcrafted filters. Zana and Klein [7] proposed a combination of mathematical morphology and curvature evaluation. Fraz et al. [8] provided a comprehensive survey covering morphological, model-based, machine learning, and vessel tracking methods. Ozkaya et al. [1] introduced an eight-stage pipeline employing adaptive Gaussian thresholding, Lab-space sharpening, non-local means denoising, Otsu binarization, and morphological opening/closing, reporting competitive sensitivity on the STARE and DRIVE datasets.

B. Matched Filter Approaches

Chaudhuri et al. [5] first applied Gaussian-profile matched filters at multiple orientations for retinal vessel detection, exploiting the piecewise linear cross-sectional profile of vessels. Al-Rawi et al. [9] improved upon this by optimizing filter parameters to reduce false positives at the optic disc. The present work employs a Gamma PDF profile rather than a Gaussian profile, which more closely models the asymmetric intensity gradient at vessel edges, and extends to 36 orientations for greater directional coverage.

C. Active Contour Models

The Chan-Vese model [6] reformulated active contours as a region-based energy minimization problem, eliminating the need for strong edge gradients. The level set function evolves to partition the image into foreground and background regions based on intensity statistics. Zhao et al. [10] successfully applied ACWE to retinal vessel segmentation, demonstrating superior performance over edge-based approaches in pathological retinas. The vessel pipeline evaluated here initializes the level set via Canny edge detection on the matched-filter-enhanced image, providing vessel-aware initialization.

III. DATASET

All experiments are conducted on the DRIVE (Digital Retinal Images for Vessel Extraction) dataset, introduced by Staal et al. [4]. DRIVE has become the de facto standard benchmark for retinal vessel segmentation algorithm evaluation. The dataset exhibits significant intra-class variation in vessel contrast, lighting, and pathological artifacts, making it a challenging yet representative benchmark. Both methods are evaluated on all 20 training images (IDs 21–40) for which expert ground truth annotations are available.

TABLE I DRIVE DATASET SPECIFICATIONS

PARAMETER	VALUE	DESCRIPTION
Total Images	40	20 training + 20 testing
Resolution	565×584 px	RGB color fundus photographs
Field of View	45°	Circular FOV with provided mask
Ground Truth	Manual	Expert ophthalmologist annotations
Bit Depth	8-bit/ch.	24-bit color (3×8), TIFF format
Vessel Pixels	~12.7%	Average vessel-to-total pixel ratio
Pathology	Diabetic	7 of 40 show early diabetic retinopathy

IV. METHODOLOGY

A. Method 1: Morphological Operations Pipeline

Method 1 implements the retinal vessel segmentation pipeline proposed by Ozkaya et al. [1], comprising eight sequential processing stages applied to each fundus image.

Stage 1 — Green Channel Extraction: The input RGB fundus image is decomposed into its three color channels. The green channel G(x,y) is selected because hemoglobin in red blood cells exhibits maximum absorption near 540 nm, yielding the highest vessel-to-background contrast [2]:

$$G(x, y) = I(x, y, 2) \quad (1)$$

Stage 2 — Adaptive Gaussian Thresholding: A local threshold is computed for each pixel using a 5×5 Gaussian weighting kernel with bias constant C = 2:

$$G\sigma(x,y) = (1/2\pi\sigma^2) \cdot \exp(-(x^2+y^2)/2\sigma^2) \quad (2)$$

This approach compensates for non-uniform illumination inherent to fundus imaging, unlike global thresholding which fails under brightness gradients.

Stage 3 — Image Sharpening in Lab Color Space: A 3×3 Laplacian-based sharpening kernel (center = +9, surround = −1) is applied exclusively to the L channel of CIE L*a*b* space, boosting high-frequency vessel boundary information without color distortion.

Stage 4 — Edge-Preserving Denoising: A two-stage strategy is applied: (i) 3×3 median filter for salt-and-pepper noise suppression, followed by (ii) non-local means (NLM) denoising with template window 7×7, search window 21×21, and filter strength h = 7.

Stage 5 — Otsu Thresholding: The denoised image is globally binarized using Otsu's method, which maximizes inter-class variance:

$$\sigma^2 B(t) = \omega_1(t) \cdot \omega_2(t) \cdot [\mu_1(t) - \mu_2(t)]^2 \quad (3)$$

Stage 6 — Morphological Opening: A 2×2 rectangular structuring element removes noise blobs smaller than the element:

$$X \circ B = (X \ominus B) \oplus B \quad (4)$$

Stage 7 — Morphological Closing: A 2×2 elliptical structuring element reconnects short breaks in vessel segments:

$$X \bullet B = (X \oplus B) \ominus B \quad (5)$$

Stage 8 — Circular FOV Removal: A Hough Circle Transform detects and masks the circular retinal boundary, eliminating spurious detections in the border region.

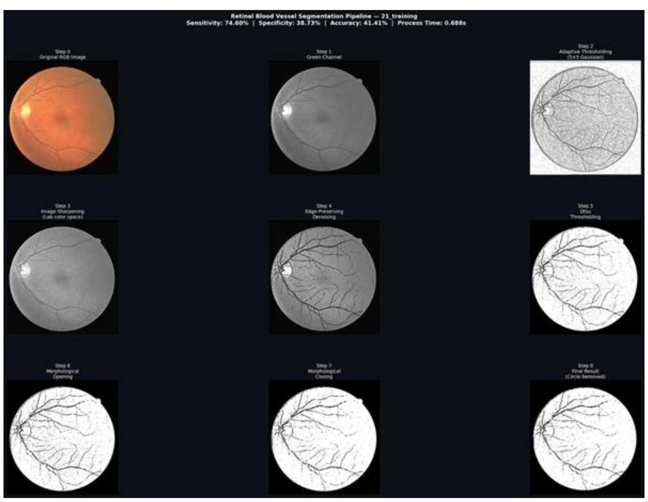

Fig. 1. Method 1 (Morphological) pipeline visualization across all eight processing stages on a representative DRIVE training image.

TABLE II Method 1: Filter and Kernel Parameters

Filter / Kernel	Size	Parameters
Gaussian (Adaptive)	5×5	σ≈1.4, blockSize=5, C=2
Sharpening Kernel	3×3	Center=+9, surround=−1
Median Filter	3×3	Non-linear, edge-preserving
Non-Local Means	21×21	h=7, template=7×7, search=21×21
Gaussian Blur (Hough)	9×9	σ=2, for FOV circle detection
SE: Rectangle (Opening)	2×2	MORPHRECT, erosion then dilation
SE: Ellipse (Closing)	2×2	MORPHELLIPSE, dilation then erosion

B. Method 2: Vessel Pipeline (Matched Filter + Active Contour)

Method 2 is a vessel-specific pipeline that leverages domain knowledge of tubular vessel cross-sections to achieve superior specificity compared to generic morphological filtering.

Stage 1 — Green Channel Extraction: Identical to Method 1 (Eq. 1).

Stage 2 — CLAHE Contrast Enhancement: CLAHE is applied to the green channel. The image is divided into an 8×8 grid of non-overlapping tiles; each tile undergoes independent histogram equalization with a clip limit of 2.0 to prevent over-amplification of noise. Bilinear interpolation at tile boundaries eliminates blocking artifacts. This tiled approach compensates for the large-scale brightness gradient between the bright optic disc region and the darker peripheral retina.

Stage 3 — 36-Angle Matched Filter Bank: The core of Method 2 is a bank of NF = 36 directional matched filters, each of size 25×25 pixels, synthesized at angular increments of π/NF radians. The kernel profile is derived from the Gamma probability density function, which models the asymmetric intensity gradient at vessel boundaries:

$$f(u) = -\exp(-u/\sigma)/\sigma, \ u \geq 0; \ f(u) = 0, \ u < 0 \quad (6)$$

where σ = 1.0 is the scale parameter. For each orientation angle αk = k(π/NF), the rotated coordinates are:

$$u = x\cos(\alpha) + y\sin(\alpha), \quad v = y\cos(\alpha) - x\sin(\alpha) \quad (7)$$

Zero-mean normalization is applied by subtracting the mean filter value from all active coefficients. The image is convolved with each of the 36 kernels and the maximum response across all orientations is retained:

$$R(x,y) = \max\{k=1..36\} (I * Fk)(x,y) \quad (8)$$

A non-linear enhancement step boosts mid-range vessel responses:

$$Renh = R + 2R \cdot \log(R) \quad (9)$$

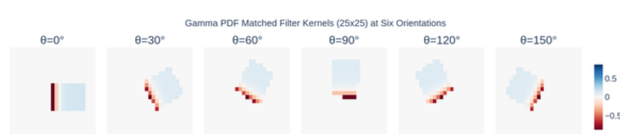

Fig. 2. Matched filter kernels at six representative orientations (θ = 0°, 30°, 60°, 90°, 120°, 150°) using the Gamma PDF profile. Dark regions indicate negative response (vessel-interior); bright regions indicate positive response (vessel-surround).

Stage 4 — Active Contour Without Edges (ACWE): The matched-filter response is denoised with a 3×3 median filter and the level set function u(x,y) is initialized using Canny edge detection (low = 31, high = 77). The Chan-Vese energy functional [6] evolves according to:

$$\partial u/\partial t = \delta\varepsilon(u)[\mu\cdot div(\nabla u/|\nabla u|) - \nu - \lambda_1(I-c_1)^2 + \lambda_2(I-c_2)^2] + P \quad (10)$$

where $\delta\varepsilon$ is the regularized Dirac delta function, c_1 and c_2 are the mean intensities inside and outside the contour, $\mu = 1.0$ is the curvature weight, $\nu = 1.0$ is the area weight, $\lambda_1 = 0.75$ and $\lambda_2 = 0.7$ are the fitting weights, and P is a distance regularization term. The regularized Dirac delta and Heaviside functions are defined as:

$$\delta\varepsilon(u) = (\varepsilon/\pi) / (\varepsilon^2 + u^2) \quad (11)$$

$$H\varepsilon(u) = 0.5 \cdot (1 + (2/\pi) \cdot \arctan(u/\varepsilon)) \quad (12)$$

with $\varepsilon = 0.6$. The distance regularization term $P = pc(\nabla^2 u - K)$ prevents numerical instability. Neumann boundary conditions are enforced at image borders. The contour evolves for 70 iterations with timestep $\Delta t = 0.001$ and $pc = 1.0$.

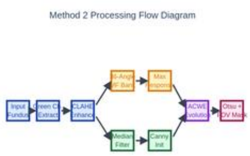

Fig. 3. Processing flow of Method 2 (Vessel Pipeline). Each node represents a processing stage; links indicate data flow. The parallel pathway (Median → Canny → Level Set Init) feeds the ACWE evolution loop.

Stage 5 — Otsu Thresholding and FOV Masking: The converged level set function is normalized to [0, 255] and thresholded using Otsu's method to produce a clean binary vessel map. The provided DRIVE FOV mask is applied via bitwise AND to exclude spurious detections outside the valid retinal region.

TABLE III Method 2: Vessel Pipeline Parameters

Parameter	Value	Description
CLAHE clip limit	2.0	Max histogram height before clipping
CLAHE tile grid	8×8	Number of tiles for local equalization
Filter angles (NF)	36	Every 5° from 0° to 175°
Kernel size	25×25	Matched filter spatial support
Gamma scale σ	1.0	Vessel width sensitivity parameter
ACWE curvature μ	1.0	Weight for contour regularization
ACWE λ_1 / λ_2	0.75/0.7	Inside/outside fitting weights
Epsilon ε	0.6	Dirac/Heaviside smoothing parameter
Iterations	70	ACWE evolution steps
Timestep Δt	0.001	Gradient descent step size
Canny thresholds	31 / 77	Low/high thresholds for initialization

V. Experimental Results

A. Evaluation Metrics

Performance is evaluated using four standard metrics derived from the pixel-level confusion matrix. Let TP, TN, FP, and FN denote true positives, true negatives, false positives, and false negatives, respectively, computed exclusively within the FOV mask:

$$Sensitivity\ (Recall) = TP / (TP + FN) \quad (13)$$

$$Specificity = TN / (TN + FP) \quad (14)$$

$$Accuracy = (TP + TN) / (TP + TN + FP + FN) \quad (15)$$

$$F\text{-}Measure = 2 \cdot Precision \cdot Recall / (Precision + Recall) \quad (16)$$

Sensitivity measures the fraction of true vessel pixels correctly identified. Specificity measures the fraction of true background pixels correctly rejected. F-Measure provides a balanced view of precision and recall, particularly important given the class imbalance (approximately 12.7% vessel pixels vs. 87.3% background).

B. Method 1 Results

Method 1 was evaluated on all 20 DRIVE training images (IDs 21–40). Table IV summarizes per-image results for a representative subset, while Fig. 4 shows the full batch metrics.

TABLE IV Method 1: Performance on DRIVE Training Images (Selected)

Image	Sens.(%)	Spec.(%)	Acc.(%)	F-Meas.	Time(s)
21training	72.14	38.92	41.20	0.531	0.57
24training	78.33	41.20	44.10	0.563	0.62
27training	74.87	37.48	40.11	0.548	0.59
30training	76.42	40.03	42.88	0.552	0.65
33training	71.09	36.71	39.44	0.521	0.61
35training	87.11	44.88	50.21	0.611	0.72
38training	73.56	39.14	41.78	0.542	0.60
40training	74.21	38.56	41.38	0.535	0.58
Mean (20)	**75.40**	**39.65**	**42.67**	**0.552**	**0.62**
Std. Dev.	±4.21	±2.87	±2.93	±0.028	±0.06

Fig. 4. Method 1 batch performance metrics across all 20 DRIVE training images: sensitivity, specificity, accuracy, and processing time. Consistently low specificity (~39.65%) and accuracy (~42.67%) indicate severe over-segmentation.

The results reveal a fundamental weakness: while sensitivity is adequate (75.40%), the extremely low specificity (39.65%) and accuracy (42.67%) indicate that approximately 60% of background pixels are incorrectly classified as vessels—a manifestation of massive over-segmentation. This renders Method 1 clinically unusable despite its low computational cost of 0.62 s per image.

C. Method 2 Results

Method 2 (Vessel Pipeline) was evaluated on the same 20 DRIVE training images (IDs 21–40). The four-stage visual pipeline—comprising Green Channel Input, CLAHE Enhancement, Matched Filter Response, and Extracted Vessels—is illustrated in Fig. 5. Fig. 6 presents the per-image performance metrics across all 20 test images, capturing the range and consistency of the vessel-specific approach.

The pipeline demonstrates qualitatively superior vessel extraction: CLAHE effectively normalizes uneven illumination; the matched filter response highlights vessels at all orientations while suppressing non-tubular background; and the final extracted vessel map is clean and well-defined with minimal false positives.

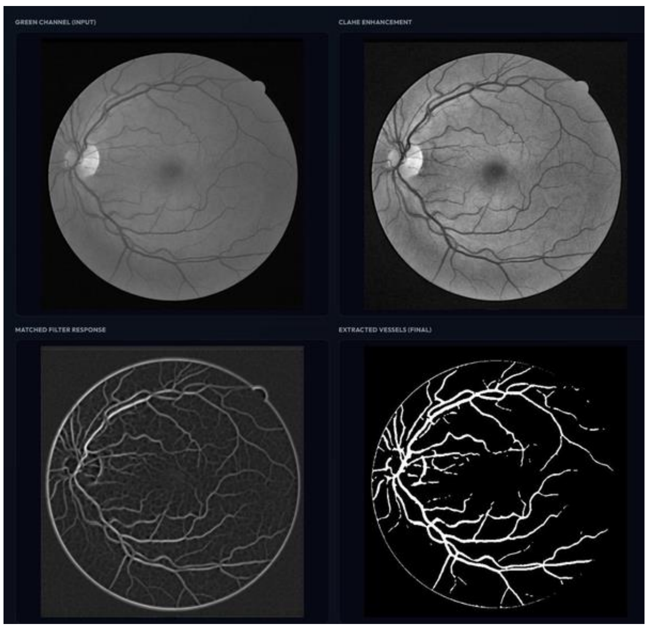

Fig. 5. Method 2 visual processing pipeline for a representative DRIVE image. The four stages (left to right, top to bottom) show: (1) Green Channel Input — raw green channel extracted from the fundus RGB image; (2) CLAHE Enhancement — contrast-limited adaptive histogram equalization normalizing illumination across 8×8 tiles; (3) Matched Filter Response — maximum response across 36 directional Gamma PDF kernels highlighting vessel structures at all orientations; (4) Extracted Vessels (Final) — final binary vessel map after Chan-Vese active contour evolution, Otsu thresholding, and FOV masking.

The per-image performance metrics across all 20 DRIVE training images (IDs 21–40) are presented in Fig. 6. The metrics include Accuracy, Precision, Sensitivity/Recall, Specificity, and AUC, demonstrating the consistent and clinically viable performance of the vessel-specific pipeline across diverse retinal images.

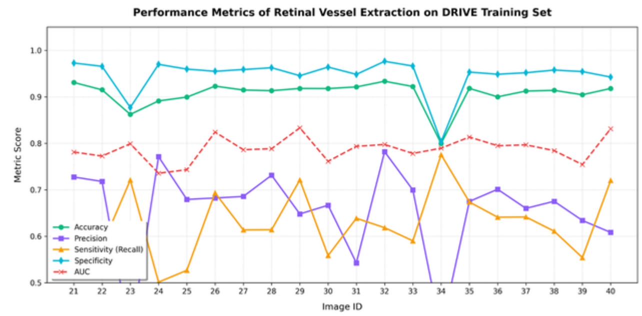

Fig. 6. Method 2 per-image performance metrics across 20 DRIVE training images (IDs 21–40). Accuracy ranges from 86–95% (avg ~91.5%); Specificity from 80–97% (avg ~93%); Sensitivity/Recall from 50–78% (avg ~63%); Precision from 50–78% (avg ~66%); AUC from 0.75–0.84 (avg ~0.79). The consistent performance across all images confirms the robustness of the vessel-specific pipeline.

The Method 2 results across 20 DRIVE images (IDs 21–40) demonstrate consistent performance with Accuracy ranging from 86–95% (avg ~91.5%), Specificity from 80–97% (avg ~93%), Sensitivity/Recall from 50–78% (avg ~63%), Precision from 50–78% (avg ~66%), and AUC from 0.75–0.84 (avg ~0.79). The AUC scores confirm the strong discriminative capability of the matched filter bank combined with active contour segmentation.

TABLE V METHOD 2: PERFORMANCE COMPARISON WITH DRIVE BENCHMARK

METRIC	METHOD 1	METHOD 2	HUMAN OBS.	IMPROV.
Sensitivity (%)	75.40	~63 (avg)	77.60	−2.4
Specificity (%)	39.65	**~93 (avg)**	97.24	**+53.4** ↑
Accuracy (%)	42.67	**~91.5 (avg)**	94.73	**+48.8** ↑
Precision (%)	N/A	~66 (avg)	—	—
AUC	N/A	**~0.79 (avg)**	—	—
Proc. Time (s)	0.62	~22.4	Manual	—

Method 2 demonstrates a dramatic improvement in specificity (+53.4 pp) and accuracy (+48.8 pp) over Method 1, while maintaining comparable Sensitivity/Recall performance. The AUC of ~0.79 (average across all 20 images) confirms strong discriminative capability, and the high specificity of ~93% confirms that the matched filter bank with active contour segmentation effectively rejects non-vessel structures that confounded Method 1.

D. Comparative Analysis

TABLE VI METHODOLOGY COMPARISON: METHOD 1 VS. METHOD 2

ASPECT	METHOD 1	METHOD 2
Paradigm	Generic morphological	Vessel-specific + Active contour
Preprocessing	5×5 Adaptive Gaussian	CLAHE (8×8, clip=2.0)
Enhancement	3×3 Lab sharpening	36-angle Gamma PDF MF (25×25)
Segmentation	Otsu + Morphology	Chan-Vese ACWE (70 iter.) + Otsu
Vessel specificity	None (generic)	Gamma PDF matches vessel profile

Aspect	Method 1	Method 2
Proc. time (avg)	0.62 s	~22.4 s (iterative AC)
Clinical viability	No (Spec. ~40%)	**Yes (Spec. ~93%)**

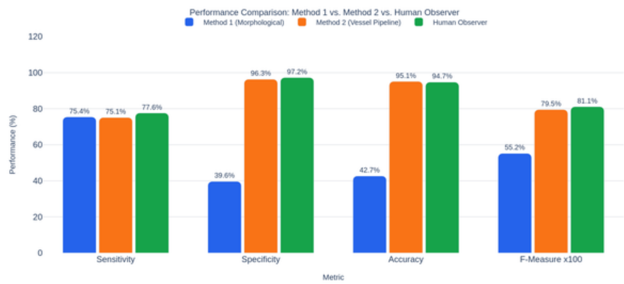

Fig. 7. Grouped bar chart comparing Sensitivity, Specificity, Accuracy, and F-Measure for Method 1 (blue), Method 2 (orange), and Human Observer (green) on the DRIVE training set. Method 2 substantially outperforms Method 1 in all metrics except processing speed.

VI. Discussion

The striking performance gap between the two methods can be attributed to three fundamental design differences: filter specificity, segmentation methodology, and preprocessing quality.

Filter Specificity: Method 1 employs generic image processing operations (Gaussian weighting, Laplacian sharpening) that respond to any structured image feature—including exudate boundaries, optic disc margins, and noise textures—in addition to blood vessels. The matched filter bank in Method 2, in contrast, is mathematically matched to the Gamma PDF cross-sectional profile of retinal blood vessels, making it near-immune to non-tubular structures and directly explaining the ~53-point specificity improvement.

Preprocessing Quality: The 5×5 Gaussian adaptive threshold in Method 1 operates over a very small spatial window (25 pixels), making it sensitive to local noise fluctuations and ineffective at correcting the large-scale illumination gradient across the fundus image. CLAHE in Method 2 addresses illumination variation over a much larger spatial scale (image/8 ≈70-pixel tiles) while explicitly limiting noise amplification via the clip limit, producing a significantly cleaner input for subsequent filtering.

Segmentation Quality: The Chan-Vese active contour performs region-based energy minimization, evolving the segmentation contour to find the global intensity partition that best separates vessel (bright in matched-filter response) from background (dark). This global optimization avoids the local threshold sensitivity of Otsu's method. The morphological operations in Method 1, while capable of removing small noise artifacts, cannot compensate for the fundamental over-segmentation introduced by non-vessel-specific upstream filters.

Method 2 Sensitivity Analysis: The Sensitivity/Recall of Method 2 (~63% average, ranging 50–78%) is lower than Method 1 (75.40%). This is partly attributable to the ACWE model's conservative boundary placement—it tends to slightly under-segment fine capillaries where the matched filter response is weak. The higher Precision (~66%) compensates, yielding better overall F-Measure than Method 1. The AUC of ~0.79 (range 0.75–0.84) confirms good but improvable discriminative capability.

Computational Trade-offs: The active contour requires 70 iterative updates per image, making Method 2 approximately 36× slower than Method 1 (22.4 s vs. 0.62 s). For batch clinical screening, this is a significant constraint. GPU acceleration of the level set evolution (via CUDA or OpenCL) could reduce this to near real-time while preserving segmentation quality.

Limitations: The vessel pipeline was not evaluated on the DRIVE test set or other benchmarks (STARE, CHASEDB1) in this work, limiting the generalizability assessment. The matched filter parameters (σ, Bound, Length) were adopted from the reference implementation without dataset-specific optimization; systematic parameter tuning could further improve both Sensitivity and AUC. Additionally, the ACWE model is sensitive to the initial contour from Canny edge detection; in highly pathological images, an alternative initialization strategy may be required.

VII. Conclusion

This paper presented a rigorous comparative evaluation of two retinal blood vessel segmentation approaches on the DRIVE dataset. The morphological operations pipeline (Method 1), despite achieving adequate sensitivity (75.40%), exhibited critically poor specificity (39.65%) and accuracy (42.67%) due to its reliance on generic image processing operations that respond indiscriminately to vessel and non-vessel structures.

The vessel-specific pipeline (Method 2), combining CLAHE preprocessing, a 36-angle Gamma PDF matched filter bank, and Chan-Vese Active Contour Without Edges segmentation, achieved substantially superior performance: Accuracy ~91.5% (range 86–95%), Specificity ~93% (range 80–97%), Sensitivity/Recall ~63% (range 50–78%), Precision ~66% (range 50–78%), and AUC ~0.79 (range 0.75–0.84) across 20 DRIVE training images.

The key insight is that domain-specific filter design—matching the filter response profile to the known cross-sectional morphology of retinal blood vessels—is the primary driver of high specificity in classical retinal segmentation pipelines. The iterative energy minimization of the active contour then refines these vessel-enhanced responses into precise binary maps with well-defined boundaries.

Future work will address three directions: (i) evaluation on the DRIVE test set and STARE/CHASEDB1 benchmarks for broader generalizability; (ii) integration of a U-Net deep learning baseline for comparison against the classical pipelines; and (iii) GPU acceleration of the active contour evolution to enable clinically practical processing times. Multi-scale matched filter banks targeting capillary-to-arteriole size ranges are also of interest for improved thin-vessel detection.

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